

JOSÉ BALTASAR LUCO KORN

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ABOUT ME

I am a dedicated Chilean student with a deep passion for physics, science, and exploring the cosmos. My lifelong aspiration is to contribute significantly to the scientific community, foster international collaborations, and engage in outreach initiatives. I possess strong teamwork abilities, a sociable demeanor, and an unyielding eagerness to continuously learn and better myself. With a structured approach to goal-setting and a proven ability to follow through, I am dedicated to ensuring the success of any project I undertake.

EDUCATION

Pontificia Universidad Católica de Chile (PUC)

March 2018-2024

B.Sc. in Astronomy (2018- 2022)

B.Sc. in Math (2021-2024)

M.Sc. in Astrophysics (2025-Present)

RESEARCH EXPERIENCE

PUC

March 2021 - Present

Research Assistant

Santiago, CHL

- Member of the research group of Professor Thomas Puzia.
- Bachelor Thesis: the Surface Brightness Fluctuation (SBF) technique on globular clusters with precision Gaia parallaxes.
- I have worked on Surface Brightness Fluctuations in Milky Way globular clusters and on the relations between SSP parameters and fluctuation colors.
- I have experience writing observing proposals, mainly for JWST and HST.
- Since 2025, I have also been studying the interstellar object 3I/ATLAS through high-resolution spectroscopy.
- In 2026, my master's thesis, "Surface Brightness Fluctuations as a Probe of Unresolved Stellar Population Parameters: A Synthetic Theoretical Framework", focuses on building composite stellar population models spanning a broad parameter space to study how SBF traces the underlying populations.

PUC

March 2019 - December 2019

Research Assistant

Santiago, CHL

- Assistant of Dr. Christopher Russell (postdoc at that moment).
- Programming a VR app (Galactic Center VR).
- Released to the public on March 24, 2020, and published on the NASA webpage.
- I collaborated with a team to develop a complimentary VR application that enables users to explore the galactic center of the Milky Way. The project was engineered using C#, Unity, and integrated data from simulations along with data from the Chandra Telescope. This experience significantly improve my programming skills and provided a profound understanding of leveraging real-world data in creating immersive virtual reality experiences.

LIST OF PAPERS, LETTERS, ARTICLES, ETC...

Luco, Baltasar; Puzia, Thomas H.; Rahatgaonkar, Rohan; et al. (2026) "Very Large Telescope observations of interstellar comet 3I/ATLAS III: High-resolution monitoring of CN and forbidden oxygen emission across the perihelion passage with ESPRESSO", *A&A*, 713, A251 (2026)
doi: 10.1051/0004-6361/202660647

Rahatgaonkar, R., Carvajal, J. P., Puzia, T. H., et al. (2025), "VLT observations of interstellar comet 3I/ATLAS II. From quiescence to glow: Dramatic rise of Ni I emission and incipient CN outgassing at large heliocentric distances", *ApJL* 995 L34
doi: 10.3847/2041-8213/ae1cbc

Puzia, T. H., Rahatgaonkar, R., Carvajal, J. P., Nayak, P. K., & Luco, B. (2025), "Spectral Characteristics of Interstellar Object 3I/ATLAS from SOAR Observations", *ApJL*, 990, L27
doi: 10.3847/2041-8213/adfa0b

Russell, C. M., Luco, B., Cuadra, J., & Sepúlveda, M. 2020, in American Astronomical Society Meeting Abstracts, Vol. 236, American Astronomical Society Meeting Abstracts #236, 324.06,
<https://ui.adsabs.harvard.edu/abs/2020AAS...23632406R>

Russell, C. M., Luco, B., Sepúlveda, M., Cuadra, J., & Calderón, D. 2021, in American Astronomical Society Meeting Abstracts, Vol. 53, American Astronomical Society Meeting Abstracts, 523.03,
<https://ui.adsabs.harvard.edu/abs/2021AAS...23752303R>

Russell, C. M. P., Luco, B., Sepulveda, M., et al. 2021, in Astronomical Society of the Pacific Conference Series, Vol. 528, New Horizons in Galactic Center Astronomy and Beyond, ed. M. Tsuboi & T. Oka, 229,
<https://ui.adsabs.harvard.edu/abs/2021ASPC..528..229R>

NASA Web page article: A New Galactic Center Adventure in Virtual Reality,
<https://www.nasa.gov/universe/a-new-galactic-center-adventure-in-virtual-reality/>

NASA Web page article Galactic Center VR: Bringing Your Neighborhood Supermassive Black Hole into Your Living Room
<https://www.nasa.gov/sc21/research/project30.html>

PRESENTATIONS

"Surface Brightness Fluctuations as Probe of Unresolved Stellar Population Parameters A synthetic Theoretical Framework", FAST-SBF in Teramo: Measuring Fast, Efficient Galaxy Distances with Surface Brightness Fluctuations, INAF-Observatory of Abruzzo/Teramo, July 2026. (Contributed Student Talk)
Link: <https://indico.ict.inaf.it/event/3480/overview>

OPEN SOURCE CODE

COMETSPEC: An open source Python module for fluorescence modeling of molecules in cometary spectra. It supports flexible MCMC fitting with freedom over multiple parameters, including mixed models of molecules, isotopologues, and elements. Currently implemented species are C₂, CN, and Fe, but the package is built to accept user-provided line lists for modeling or fitting. Check the documentation available at: <https://cometspec.readthedocs.io/en/latest/>

Interactive Spectra Viewer: An online tool for quickly displaying spectra. Its main feature is point decimation, rendering fewer points for faster display of large spectra. It supports multiple file formats and allows plotting different columns. You can find it at: https://baltasarluco.github.io/pages/spectral_viewer.html

WORK HISTORY

PUC

Outreach and Diffusion assistant

April 2022 - January 2023

Santiago, CHL

- As an Outreach Assistant at the Astrophysics Institute of Pontificia Universidad Católica de Chile, I refined my skills in science communication through multimedia editing, social media management, live streaming, and promoting astronomy education. Additionally, I was entrusted with the role of Monitor Coordinator, orchestrating the public opening of the historical Manuel Foster Observatory, a recognized National Monument. My efforts contributed to the accessible dissemination of astronomical knowledge and the preservation of our astronomical heritage.

Center of Excellence in Astrophysics and Related Technologies (CATA)

April 2022 - December 2022

*Outreach and Diffusion assistant**Santiago, CHL*

- As an Outreach Assistant at CATA, I engaged in science communication initiatives, multimedia editing, social media management, and the dissemination of astrophysics knowledge. Additionally, I coordinated and executed a diverse range of educational outreach activities across various schools and provinces within Santiago, promoting scientific literacy and fostering community engagement with astrophysics.

PUC

August 2024 - December 2024

*Professor's Assistant**Santiago, CHL*

- Assistant to Prof. Jorge González in the Experimental Astrophysics class

PUC

August 2022 - December 2022

*Professor's Assistant**Santiago, CHL*

- Assistant to Prof. Felipe Barrientos in the Experimental Astrophysics class

PUC

March 2021 - December 2021

*Professor's Assistant**Santiago, CHL*

- Assistant to Prof. Thomas Puzia in the Astronomy class, a university-wide course open to all majors.
- 2 semesters

Vail Resorts

December 2019 - March 2020

*Ski Instructor**Breckenridge, CO, USA*

- I traveled to Colorado to work as a ski instructor under the J1 Visa program, specializing in teaching children aged 3 to 6 years old. This role provided a valuable opportunity to enhance my teamwork skills, foster social relationships, gain experience in working with children, and embody a strong sense of responsibility.

OUTREACH

PUC

2022

*Tour Guide**Santiago, CHL*

- Served as a tour guide during the Heritage Cultural Day at the Manuel Foster Observatory and throughout the year, enriching visitors' experiences by sharing the site's historical and cultural significance.

PUC

2022

*Livestream**Santiago, CHL*

- Engaged in the illustrious PUC livestream coverage of the 2022 Partial Solar and Total Lunar Eclipses.

PUC-CATA

2022

*Social Media**Santiago, CHL*

- Producing a spectrum of informative to educational content for CATA and PUC's social media outlets (Instagram and Facebook).

CATA

2022

*Kid Workshops**Santiago, CHL*

- Presenter at various workshops for children on comets and asteroids during my engagement at CATA.

PUC

2019

*Diffusion**La Serena, CHL*

- During the 2019 Total Eclipse, our VR team showcased our app at Mamalluca's Observatory and Vicuña Town Hall, Chile, providing insights and educating individuals of all ages about the center of the Milky Way.

SCHOLARSHIPS AND HONOURS

PUC

March 2018 - December 2018

- Academic Excellence Award (Honors) in the Astronomy Bachelor

VOLUNTEERING

TECHO

August 2020 - March 2022

Voluntier

Santiago, CHL

- I was an active member of the Chilean TECHO foundation team from August 2020 to March 2022, focusing on housing solutions for a community in La Florida, Chile. My involvement spanned across fostering social relations, aiding in construction projects, and demonstrating a robust commitment to enhancing living conditions. This experience sharpened my interpersonal skills, provided practical construction insights, and fostered a deep sense of community engagement, all of which significantly enriched my perspective and approach towards social housing solutions.

San Isidro School

2017

Voluntier

Santiago, CHL

- As part of a volunteer team, I visited a senior care hospital where we assisted with the care, feeding, and cleaning for the elderly residents. Additionally, we helped maintain the cleanliness of the facility. Throughout this experience, we fostered meaningful relationships with the patients, enhancing their social interactions and overall well-being.

COURSEWORK

NAME	CODE COURSE	GRADE	LEVEL
COSMOLOGY AND DARK MATTER	ASP5413	6.7 / 7 (3.88 / 4)	GRADUATE
ADVANCE EXTRAGALACTIC ASTROPHYSICS	ASP5302	6.6 / 7 (3.84 / 4)	GRADUATE
STELLAR POPULATIONS	ASP5106	6.6 / 7 (3.84 / 4)	GRADUATE
PHYSICAL PROCESS IN ASTROPHYSICS	ASP9110	6.7 / 7 (3.88 / 4)	GRADUATE
INTRODUCTION TO ASTROCHEMISTRY	ASP5555	6.5 / 7 (3.80 / 4)	GRADUATE
GENERAL ASTROPHYSICS	AST0311	6.9 / 7 (3.96 / 4)	UNDERGRADUATE
STELLAR ASTROPHYSICS	AST0321	7 / 7 (4 / 4)	UNDERGRADUATE
EXTRAGALACTIC ASTROPHYSICS	AST0411	6.7 / 7 (3.88 / 4)	UNDERGRADUATE
EXPERIMENTAL ASTROPHYSICS.	AST0421	6.4 / 7 (3.76 / 4)	UNDERGRADUATE
SUPER MASSIVE BLACK HOLES.	AST0311	6.9 / 7 (3.96 / 4)	GRADUATE
RELATIVITY AND GRAVITATION	ASP5412	5.6 / 7 (3.36 / 4)	GRADUATE
VARIABLE STARS	ASP5203	6.9 / 7 (3.96 / 4)	GRADUATE
ADVANCE PROGRAMMING	IIC2233	5.9 / 7 (3.54 / 4)	UNDERGRADUATE
DIFFERENTIAL GEOMETRY	MAT2305	5.8 / 7 (3.48 / 4)	UNDERGRADUATE
QUANTUM PHYSICS II	FIZ0412	7/7 (4/4)	UNDERGRADUATE
AMONG OTHERS			

STRENGTHS

Computer Languages	Python, C#
Tools	Latex, Unity, Pycharm
Favorites packages	Numpy, Matplotlib, Scipy, Pandas, Astropy
Multimedia	Davinci Resolve, Pixelmator, GIMP
Languages	Spanish (native), English (fluent)

INTERESTS AND HOBBIES

Interests	Stellar Populations, Galaxy Evolution, Cosmology, Relativity, Theoretical Astrophysics
Hobbies	Photography, Gardening, Coffee, Homebrewing, Writing, Reading, Poetry, Skiing